Metric Spaces

Let $X \neq \emptyset$; a <u>distance</u> on X is a map $d : X \times X \to [0, \infty)$ such that

1) $d(x,y) \geq 0$, $\forall (x,y)$
2) $d(x,y) = 0 \iff x = y$
3) $d(x,y) = d(y,x)$
4) $d(x,z) \leq d(x,y) + d(y,z) \quad \forall x,y,z \in X$

A <u>metric space</u> is a set $X \neq \emptyset$ with a distance (metric) : (X, d)

<u>Examples</u>

1) $(\mathbb{R}, d) \quad d(x,y) = |x-y|$
2) $(\mathbb{Q}, d) \quad d(x,y) = |x-y|$
3) $(\mathbb{C}, d) \quad d(x,y) = |x-y|$
4) $(\mathbb{R}^n, d)$; $d(x,y) = \sqrt{\sum_{j=1}^{n} (x_j - y_j)^2}$

$x = (x_1 \dots x_n)$

$y = (y_1 \dots y_n)$

5) $(C[a,b], d_1) \quad d_1(f,g) = \int_a^b |f(x) - g(x)| \, dx$
6) $(C[a,b], d_\infty) \quad d_\infty(f,g) = \max_{x \in [a,b]} |f(x) - g(x)|$

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Def **Notation** Let (X, d) be a metric space and $a \in X$, $r > 0$

The open disk of center a and radius r

$$B(a,r) = \{x \in X \mid d(x,a) < r\}$$

The closed disk $\overline{B(a,r)} = \{x \in X \mid d(x,a) \leq r\}$

The **Sphere (circle)** $S(a,r) = \{x \in X \mid d(x,a) = r\}$

Examples : $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{C}$, $\mathbb{C}[a,b]$

Def **Open set** Let (X, d) be a metric space

$D \subset X$ is called an **open set** if

$\forall a \in D$, $\exists r > 0$ s.t. $B(a,r) \subset D$

Closed set $F \subset X$ is a **closed set**

if $X \setminus F$ is open.

Bounded set $A \subseteq X$ is a **bounded set** if $\exists a \in X$ and $r > 0$:

$$A \subseteq B(a,r)$$

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Sequences in metric spaces

Def Let (X, d) be a metric space

A sequence in X is $f: \mathbb{N} \to X$

Notation: $f(n) = x_n$, $n \in \mathbb{N}$

Def x_n is bounded if $\exists\, a \in X, r > 0$ such that $x_n \in B(a, r)$, $\forall n \in \mathbb{N}$

Def (Convergent Sequence)

Let $x_n \in X$; x_n is called convergent

iff $\exists\, a \in X$ such that $\forall \varepsilon > 0$, $\exists N_\varepsilon \in \mathbb{N}$

with: $d(x_n, a) < \varepsilon$, $\forall n \geq N_\varepsilon$

a = the limit of x_n: $x_n \to a$

Obs a is unique (if it exists):

$$d(a, a') \leq d(a, x_n) + d(x_n, a')$$

Examples: in $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{R}^n$, $(\mathcal{C}[a,b], d_\infty)$

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Def (Cauchy sequence) Let $x_n \in X$ (X metric space). x_n is called a Cauchy sequence iff

$\forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N} : d(x_n, x_m) < \varepsilon,\ \forall n, m \geq N_\varepsilon$

Obs x_n convergent $\Rightarrow$ x_n Cauchy seq

Not converse?

In general it is not true:

$(\mathbb{Q}, d)$ $\quad d(x,y) = |x-y|$

$x_n = \left(1 + \frac{1}{n}\right)^n \longrightarrow$ Cauchy

$\longrightarrow$ divergent

Def (X, d) is a **complete metric space** iff: x_n Cauchy seq $\Rightarrow$ x_n convergent

Ex: $\mathbb{R}, \mathbb{C}, \mathbb{R}^n, (\mathcal{C}[a,b], d_\infty)$

$([a,b], |x-y|), [a, \infty), (-\infty, a]$

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The Contraction Theorem
(Fixed Point principle)
the Method of successive approximations

Let (X,d) be a metric space and

$f : X \to X$

$a \in X$ is called a fixed point of f if

$$\boxed{f(a) = a}$$

Def Let (X,d) and $T : X \to X$.

T is a contraction (on X) if $\exists k \in (0,1)$ such that $d(T(x), T(y)) \leq k\, d(x,y)$, $\forall x,y \in X$

k = the contraction factor of T

The Contraction Theorem

Let (X,d) be a complete metric space and $T : X \to X$ a contraction.

Then there exists a unique $\xi \in X$ such that $T(\xi) = \xi$.

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On a complete metric space, a contraction has a unique fixed point.

Proof (the sequence of successive approximations)

Let $x_0 \in X$ and let $x_1 = T(x_0)$,

$x_2 = T(x_1)$, ... $\boxed{x_n = T(x_{n-1})}$

We now prove that x_n is Cauchy:

$d(x_{n+1}, x_n) = d(T(x_n), T(x_{n-1})) \leq$

$\leq k\, d(x_n, x_{n-1}) \leq k\, d(T(x_{n-1}), T(x_{n-2})) \leq$

$\leq \dots \leq k^n d(x_1, x_0)$

Hence $\boxed{d(x_{n+1}, x_n) \leq k^n d(x_1, x_0)}$

$d(x_{n+p}, x_n) \leq d(x_{n+p}, x_{n+p-1}) + d(x_{n+p-1}, x_n) \leq$

$\dots \leq k^{n+p-1} d(x_1, x_0) + \dots + k^n d(x_1, x_0) =$

$= k^n (1 + k + k^2 + \dots + k^{p-1})\, d(x_1, x_0) \leq$

$\leq \frac{k^n}{1-k}\, d(x_1, x_0) \quad \forall n \in \mathbb{N}$

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Hence: $\boxed{d(x_{n+p}, x_n) \leq \frac{k^n}{1-k} d(x_1, x_0)}$

It results that (x_n) is Cauchy, hence x_n is convergent (X is complete)

Let $\xi = \lim_{n\to\infty} x_n \in X$

Then $T(\xi)$ (obviously) and ξ is unique: $d(\xi, \xi') \leq k\, d(\xi, \xi')$ —

Moreover:

The error formula

$$\boxed{d(x_n, \xi) \leq \frac{k^n}{1-k} d(x_1, x_0)}$$

The sequence x_n = the sq of successive approximations.

Observations: (The limit doesn't depend on x_0.)

Applications

1) If $f : [a,b] \to \mathbb{R}$, $\mathcal{C}^1 \Longrightarrow$

$\exists\, c \in (a,b)$ $|f(b) - f(a)| \leq |f'(c)|\,|b-a|$

It results that if

$\max_{x \in [a,b]} |f'(x)| = k \in (0,1)$, then

f is a contraction.

Equations Approximate the solution of $x^3 + 4x - 1 = 0$ with an error $\varepsilon < 10^{-3}$.

$\xi \in (0,1)$

$x(x^2+4) = 1 \iff x = \dfrac{1}{x^2+4} = T(x)$

$T : [0,1] \to [0,1]$ is a contraction:

$T'(x) = \dfrac{-2x}{(x^2+4)^2}$, $|T'(x)| = \dfrac{2x}{(x^2+4)^2}$

$\max_{x \in [0,1]} |T'(x)| = \dfrac{2}{25}$ (exercise!!)

Hence T is a contraction with factor $k = \dfrac{2}{25}$

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Cat $x_0 = 0 \in [0,1]$, $x_1 = T(0) = \frac{1}{4}$

$x_n = T(x_{n-1}) \Rightarrow \boxed{x_n = \frac{1}{x_{n-1}^2 + 4}}$

$d(x_n, \xi) \leq \frac{k^n}{1-k} d(x_1, x_0)$, de unde

$|x_n - \xi| \leq \frac{\left(\frac{2}{25}\right)^n}{1 - \frac{2}{25}} \left|\frac{1}{4}\right| = 1$

$\Rightarrow |x_n - \xi| \leq \left(\frac{2}{25}\right)^n \cdot \frac{25}{92} < \varepsilon = 10^{-3}$

$\Rightarrow n = 3 \Rightarrow \xi \approx x_3 = T(T(\frac{1}{4})) =$

$= T\left(\frac{16}{65}\right) \approx 0,2462$

2) Fractals:

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